BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 319 · Paper 2 targeted reteaching

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is O(log n)?

Q2. What is a recursive base case?

Q3. What does a trace table show?

Q4. Trace search for 3 in sorted [1,3,5,7,9].

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I Do · Read new knowledge

Key vocabulary

bounds — Limits on a search interval.

base case — Recursive stopping condition.

auxiliary space — Extra storage beyond input.

New learning

Identify — Locate incorrect condition or state update

Model — Model a correct trace table

Apply — Test new data, including missing target

Evaluate — Explain time and memory consequences

Worked example

Trace search for 3 in [1,3,5,7,9].

1. Explain the method — The midpoint is excluded after a failed comparison, so bounds shrink.

We Do · Practise with guidance

Repair recursive SumTo for n=2.

```
if(n==0)return 0; else return n+SumTo(n-1);
```

Explain the method

SumTo(2)=2+SumTo(1)=2+1+SumTo(0)=3.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Trace binary search for 4 in [2,4,6,8,10], giving bounds/midpoints. [6]

Question 2

Trace recursive SumTo(4), including base0 and return values. [6]

Question 3

Compare bubble and merge sort worst-case time/extra space; choose for million values with sufficient memory. [6]

Exit Ticket · Answer before checking

Binary search may never update the bounds but still always terminate.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Work growing with the number of halvings.

Q2. Condition stopping further calls.

Q3. Intermediate program states.

Q4. mid2→5, high1, mid0→1, low1, mid1→3; return index1.

You Do pages · Independent questions

Q1. mid2→6 [2], high1 [1], mid0→2 [1], low1 then mid1→4 [1], return index1 [1].

Q2. Calls 4,3,2,1,0 [3], returns 0,1,3,6,10 [3].

Q3. Bubble $O(n^2)$ [1] and $O(1)$ extra [1]; merge $O(n \log n)$ [1] and $O(n)$ extra [1]; merge scales better [1] with available memory [1].

Exit Ticket · Model answer

False. It can repeat the same midpoint.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: